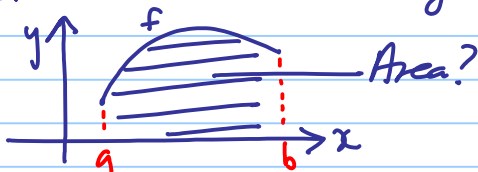


Suppose we have the following situation:



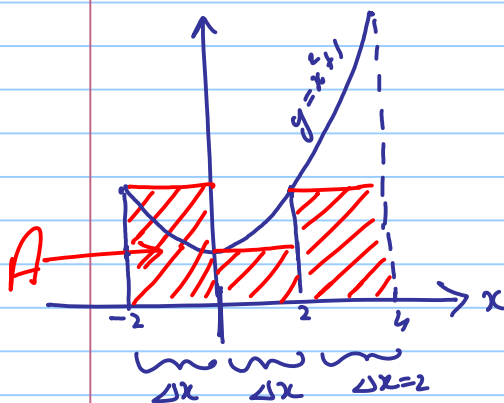
We have a function f , continuous on an interval $[a, b]$, and $f(x) > 0$ on $[a, b]$. How do we calculate (or estimate) the area between the graph, the x -axis and the lines $x=a$ and $x=b$?

- ① First of all we will use left (or right) endpoints of subintervals to estimate the area.

Examples:

Find the area under the parabola $f(x) = x^2 + 1$ on $[-2, 4]$

- (a) Use ⁿ 3 subintervals and the left endpoint of the sub-intervals.



We have 3 subintervals:

$(-2, 0)$, $(0, 2)$ and $(2, 4)$

The length of each subinterval is $\Delta x = \frac{b-a}{n} = \frac{4-(-2)}{3} = 2$.

When using a left-hand sum, we will use the function values at the left endpoints of the

sub-intervals: $x = -2, x = 0$ and $x = 2$

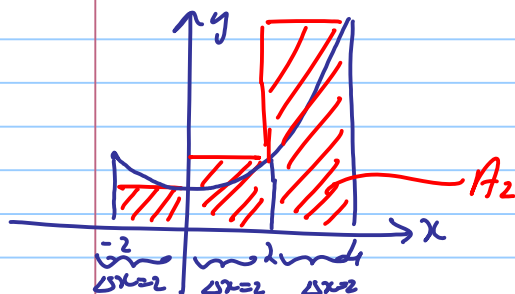
$$\begin{aligned} \Rightarrow A &= [f(-2) + f(0) + f(2)] \Delta x \\ &= [5 + 1 + 5] \times 2 \\ &= 11 \times 2 = 22 \end{aligned}$$

$$f(-2) = (-2)^2 + 1 = 5$$

$$f(0) = 0^2 + 1 = 1$$

$$f(2) = 2^2 + 1 = 5$$

Now estimate the same area using the right endpoints of the subintervals, which are $x=0, x=2, x=4$



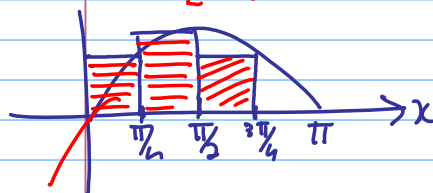
$$f(0) = 1$$

$$f(2) = 5$$

$$f(4) = 4^2 + 1 = 17$$

$$\begin{aligned} A_2 &= [f(0) + f(2) + f(4)] \Delta x \\ &= [1 + 5 + 17] \times 2 = 46 \end{aligned}$$

- (2) Find the area under the graph of $y = \sin x$ on $[0, \pi]$, using the right endpoints of 4 subintervals.
 $[a, b]$



In this case: $\Delta x = \frac{\pi - 0}{4} = \frac{\pi}{4}$

Subintervals are: $(0, \frac{\pi}{4})$, $(\frac{\pi}{4}, \frac{\pi}{2})$, $(\frac{\pi}{2}, \frac{3\pi}{4})$, $(\frac{3\pi}{4}, \pi)$

The function values at the right endpoints are:

$$f(\frac{\pi}{4}) = \frac{1}{\sqrt{2}}, f(\frac{\pi}{2}) = 1, f(\frac{3\pi}{4}) = \frac{1}{\sqrt{2}}, f(\pi) = 0.$$

$$\begin{aligned} \Rightarrow A_3 &= [f(\frac{\pi}{4}) + f(\frac{\pi}{2}) + f(\frac{3\pi}{4}) + f(\pi)] \Delta x \\ &= [\frac{1}{\sqrt{2}} + 1 + \frac{1}{\sqrt{2}} + 0] \times \frac{\pi}{4} \\ &= [1 + \sqrt{2}] \frac{\pi}{4}. \end{aligned}$$

- (3) Over-estimate and Under-estimate (Increasing/decreasing functions)

Consider an increasing function:

e.g. $y = x^2 + 1$ on $[0, 2] = [a, b]$

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{2} = 1$$

Find the area under the graph using
 (1) left hand end-points of 2 sub-intervals

$$f(0) = 0 + 1 = 1$$

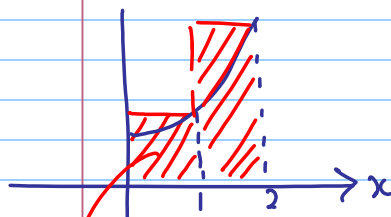
$$f(1) = 1 + 1 = 2$$



$$\begin{aligned} A_4 &= [f(0) + f(1)] \Delta x \\ &= [1 + 2] \times 1 \\ &= 3 \end{aligned}$$

A_4 is an under-estimate of the area underneath $y = x^2 + 1$ on $[0, 2]$.

- (2) Right hand endpoints and 2 sub-intervals:



The 2 subintervals are:

$$(0, 1) \text{ and } (1, 2)$$

$$\Rightarrow \Delta x = 1.$$

$$f(1) = (1)^2 + 1 = 2$$

$$f(2) = (2)^2 + 1 = 5$$

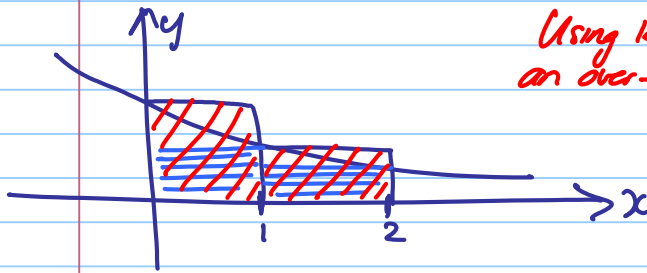
$$\begin{aligned} A_5 &= [f(1) + f(2)] \Delta x \\ &= [2 + 5] \times 1 = 7 \end{aligned}$$

A_5 is an over-estimate of the area

under the graph of $f(x) = x^2 + 1$, between $x=0$ and $x=2$.

For a decreasing function:

$f(x) = e^{-x}$ on $[0, 2]$, using 2 rectangles, we find:



Using left endpoints gives us an over-estimate of the area.

Using right endpoints gives us an under-estimate of the area.